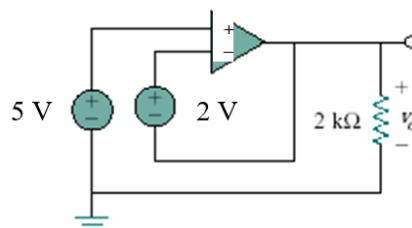
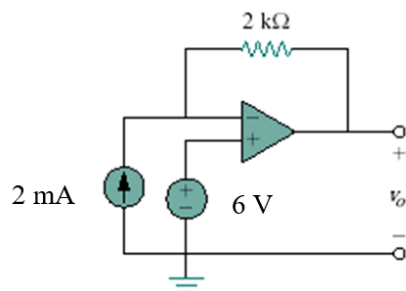
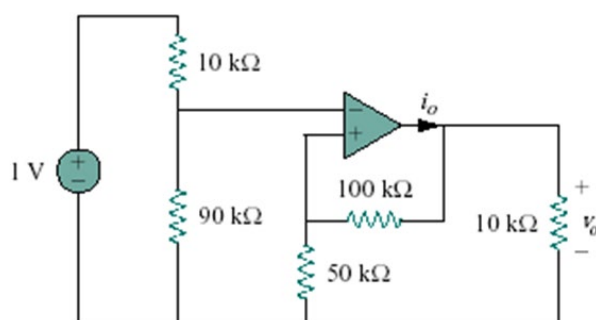


HW 5_Ch5

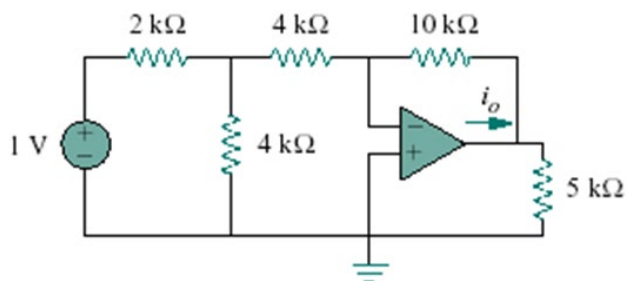
- 1) Determine v_o for each of the op amp circuits below.



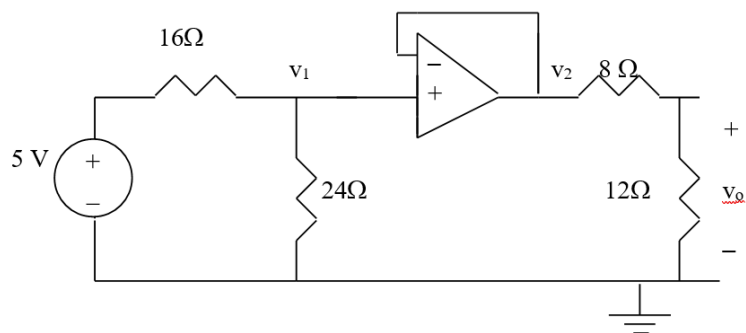
- 2) Find v_o and i_o in the circuit below.



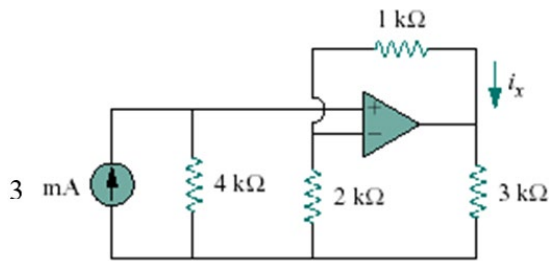
- 3) Determine i_o in the circuit below.



- 4) Find v_o in the op amp circuit below.



- 5) Refer to the op amp circuit below. Calculate i_x and the power dissipated by the 3-k Ω resistor.

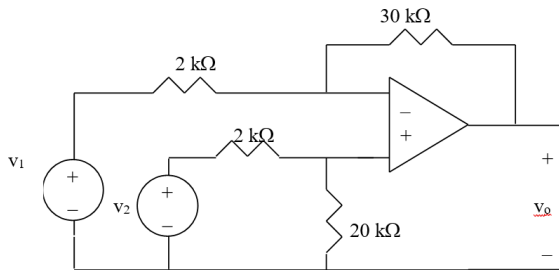


- 6) An *averaging amplifier* is a summer that provides an output equal to the average of the inputs. By using proper input and feedback resistor values, one can get

$$-v_{out} = \frac{1}{4}(v_1 + v_2 + v_3 + v_4)$$

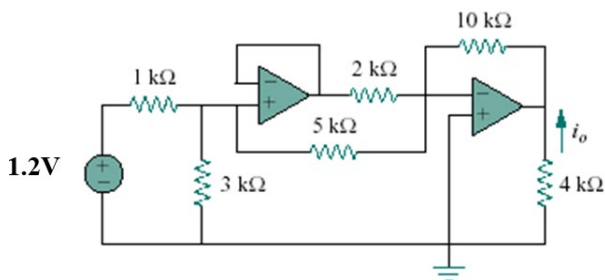
Using a feedback resistor of 10 k Ω , design an averaging amplifier with four inputs.

- 7) The circuit below is for a difference amplifier. Find v_o given that $v_1 = 1V$ and $v_2 = 2V$.



- 8) Design an op amp circuit such that $v_o = -2v_1 + 4v_2 - 5v_3 - v_4$
Let all the resistors be in the range of 5 to 100 k Ω .

- 9) Calculate i_o in the op amp circuit below.



- 10) A noninverting current amplifier is below. Calculate the gain i_o/i_s . Take $R_1 = 8$ k Ω and $R_2 = 1$ k Ω .

